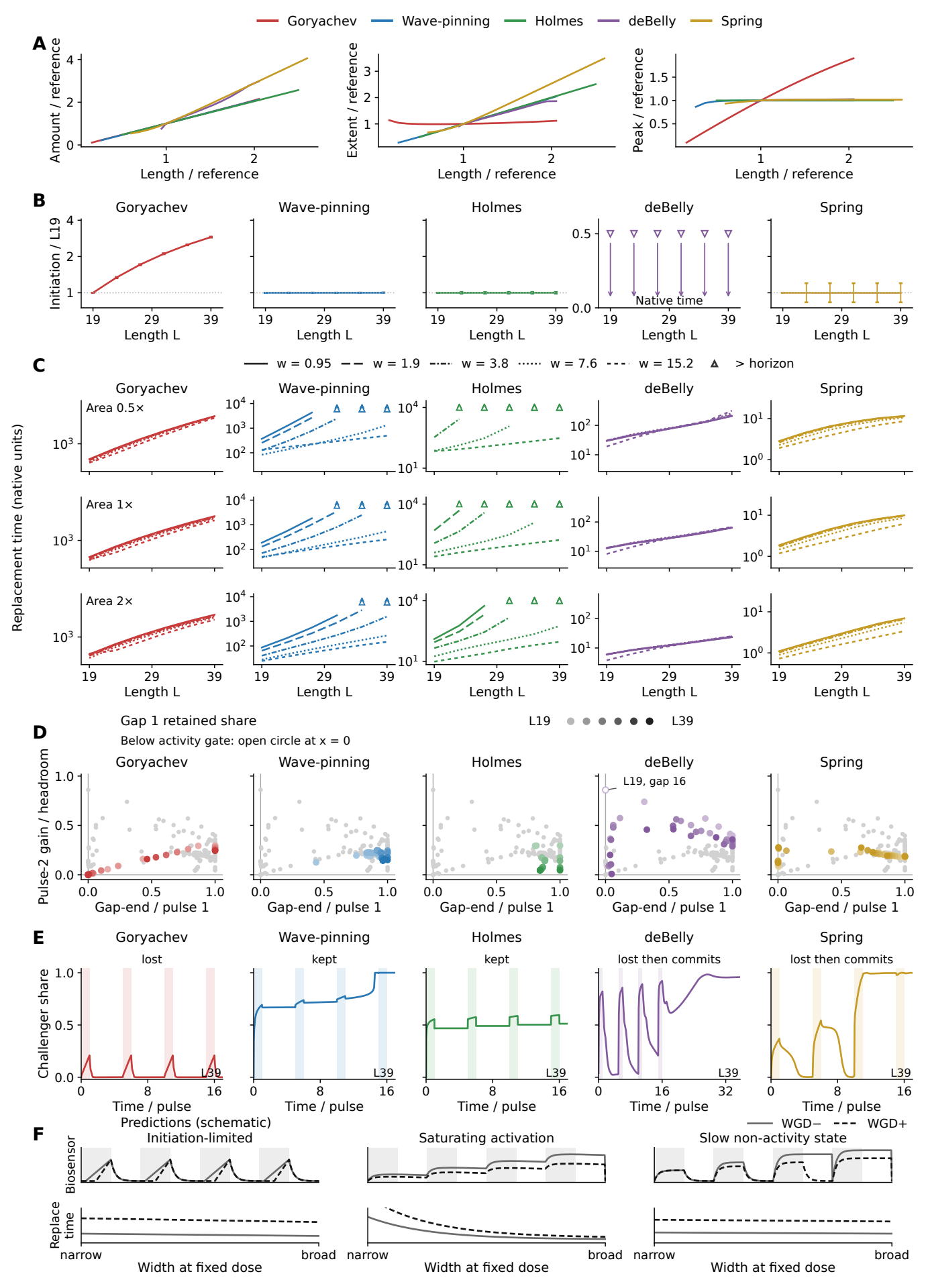
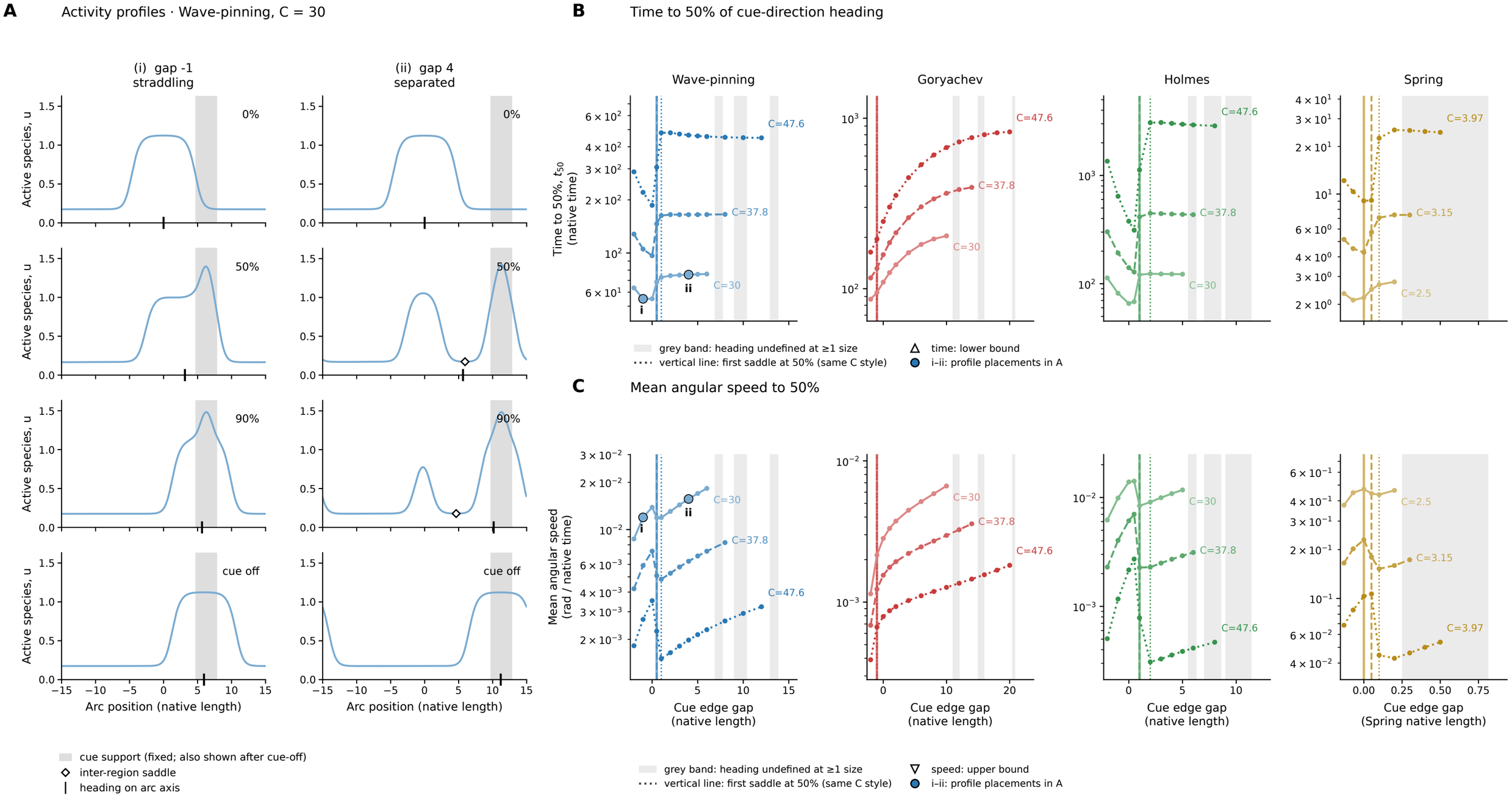

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose


A
Activity profiles · Wave-pinning, C = 30
(i) gap -1
straddling
(ii) gap 4
separated
0%
50%
90%
cue off
Active species, u
Arc position (native length)
cue support (fixed; also shown after cue-off)
inter-region saddle
heading on arc axis
B
Time to 50% of cue-direction heading
Wave-pinning
Goryachev
Holmes
Spring
Time to 50%, t_{50}
(native time)
C=47.6
C=37.8
C=30
C=3.97
C=3.15
C=2.5
i
ii
grey band: heading undefined at ≥1 size
vertical line: first saddle at 50% (same C style)
time: lower bound
i–ii: profile placements in A
C
Mean angular speed to 50%
Mean angular speed
(rad / native time)
Cue edge gap
(native length)
Cue edge gap
(Spring native length)
speed: upper bound

A

Drifting peak (same physical D_s) | Jumping peak (same rate) | Jumping peak (rate ∝ perimeter)

Time (×10³) — C20 — C60

Heading $C(\ell)$ — Lag (×10³)

$\alpha = 2$ (C = 20, C = 60) | $\alpha = 0$ (curves coincide) | $\alpha = -1$

B

C: 20 30 40 50 60

Drift only | Drift + same-rate jumps | Drift + perimeter-rate jumps

Local exponent α — Input fraction f

drift | same-rate jumps | perimeter-rate jumps

$f_{1/2}/f_{1/2}(C_{20})$ — Circumference C

Drift | + same-rate jumps | + perimeter-rate jumps

$\propto C$

C

Wave-pinning: 20 30 40 50 60 | Holmes: 20 30 40 50 60 | Spring: 1 1.5 2 2.5 3 | Goryachev: 20 30 40 50 60

C; Spring ×20 in fold panel

Wave-pinning | Holmes | Spring | Goryachev

Local exponent α — Input fraction f

$f_{1/2}/f_{1/2}(C_{20})$ — Circumference C

Wave-pinning | Holmes | Spring | Goryachev

$\propto C$

D

Wave-pinning, $f = 0.03$ | Holmes, $f = 0.02$ | Spring, $f = 0.005$ | Goryachev, $f = 0.01$

Time (×10³) — switch — C20 — C60 (C1 — C3 for Spring)

Heading $C(\ell)$ — Lag (×10³)

$\alpha_{20,60} = 2.1$, 95% CI [1.9, 2.3] (C = 20, C = 60)

$\alpha_{20,60} = 1.4$, 95% CI [1.3, 1.5]

$\alpha_{20,60} = 1.8$, 95% CI [1.7, 1.9]

$\alpha_{20,60}$ unavailable

E

Wave-pinning | Holmes | Spring | Goryachev

$D_s/D_s(C_{\min})$

λ_{eff} — Scored events — Mixture fit

$k_{\text{pred}}/k_{\text{obs}}$

Multiple-front fraction

Input fraction f